

Problem Set 3:

Submission: 12:00, Fri, May 01, 2026; Solutions: Fri, May 15, 2026

Problem 1 (Zentralübung) Multipole Expansion of an Uniformly Charged Disk

Let us return to the scenario from exercise 3 in last week's problem set: Consider a two-dimensional disk of radius R with a uniform charge per unit area $\sigma > 0$. For concreteness, choose the origin to coincide with the center of the disk, and orient it such that the disk lies in the $x - y$ -plane. Now, we want to recover the solution we obtained for the electrostatic potential $\phi(z)$ on the z -axis by other, more general, means.

(1.a) Start from the *exterior multipole expansion* of the electrostatic potential,

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} A_{\ell m} \frac{Y_{\ell m}(\phi, \theta)}{r^{\ell+1}}, \quad r > R, \quad (1)$$

(see also Eq. (2.106) in the lecture script) to show that the potential of our disk can be expanded in the following form:

$$\phi(r, \theta) = \frac{Q}{4\pi\epsilon_0 r} \sum_{\ell=0}^{\infty} \left(\frac{R}{r}\right)^{\ell} \frac{2}{\ell+2} P_{\ell}(0) P_{\ell}(\cos \theta), \quad r > R. \quad (2)$$

To make connection between Eqs. (1) and (2), compute the *exterior spherical multipole moments* $A_{\ell m}$ of the charge distribution:

$$A_{\ell m} = \frac{4\pi}{2\ell+1} \int d^3\mathbf{r}' \rho(\mathbf{r}') r'^{\ell} Y_{\ell m}^*(\phi', \theta'). \quad (3)$$

(1.b) Use your result from part (a) to confirm that your result for the potential $\phi(z)$ on the z -axis agrees with the answer you got in last week's problem set when $z > R$.

Hint 1: $P_{\ell}(1) = 1$

Hint 2: The following identity involving the generating function of the Legendre polynomials may be useful:

$$\frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x) t^n \quad (4)$$

(1.c) Using Hilb's formula,

$$P_{\ell}(\cos(\theta)) = \sqrt{\frac{2}{\pi\ell \sin(\theta)}} \cos\left[\left(\ell + \frac{1}{2}\right)\theta - \frac{\pi}{4}\right] + \mathcal{O}(\ell^{-3/2}), \quad \theta \in (0, \pi), \quad (5)$$

show that the in-plane electric field diverges logarithmically as $r \downarrow R$.

- (1.d) (*Optional*) Use your result from part (a) to plot $\phi(r)$ (e.g. using Python) and the electric field $\mathbf{E} = -\nabla\phi$ in the x - y -plane for $r > R$ and check whether it agrees with the sketch you made last week.

Solution:

- (1.a) To compute the exterior spherical multipole moment A_{lm} of the charge distribution

$$A_{lm} = \frac{4\pi}{2\ell + 1} \int d^3\mathbf{r} \rho(\mathbf{r}) r^\ell Y_{\ell m}^*(\phi, \theta), \quad (6)$$

we first have to parametrize the charge distribution in terms of spherical coordinates

$$d^3\mathbf{r} \rho(\mathbf{r}) = d^3\mathbf{r} \delta(z) \Theta(R - \sqrt{x^2 + y^2}) \frac{Q}{\pi R^2} \quad (7)$$

$$= d^3\mathbf{r} \delta(r \cos(\theta)) \Theta(R - r \sin(\theta)) \frac{Q}{\pi R^2} \quad (8)$$

$$= d^3\mathbf{r} \left(\frac{\delta(\theta - \pi/2)}{r |\sin(\theta)|} + \frac{\delta(r)}{|\cos(\theta)|} \right) \Theta(R - r \sin(\theta)) \frac{Q}{\pi R^2} \quad (9)$$

$$= d^3\mathbf{r} \frac{1}{r} \delta(\theta - \pi/2) \Theta(R - r) \frac{Q}{\pi R^2}, \quad (10)$$

where Θ is the Heavyside Theta function. In the last step, we have used that with the integration measure $d^3\mathbf{r} = r^2 \sin(\theta) dr d\phi d\theta$, the second term with $\delta(r)$ vanishes. With this, we obtain

$$A_{lm} = \frac{4\pi}{2\ell + 1} \int d^3\mathbf{r} \rho(\mathbf{r}) r^\ell Y_{\ell m}^*(\phi, \theta) \quad (11)$$

$$= \frac{4\pi}{2\ell + 1} \frac{Q}{\pi R^2} \int_0^R dr r^{\ell+1} \int_0^{2\pi} d\phi Y_{\ell m}^*(\phi, \theta = \pi/2). \quad (12)$$

Now, the integral of the radius r becomes

$$\int_0^R dr r^{\ell+1} = \frac{R^{\ell+2}}{\ell + 2}, \quad (13)$$

and the the integral over the angle ϕ becomes

$$\int_0^{2\pi} d\phi Y_{\ell m}^*(\phi, \pi/2) = 2\pi \delta_{m,0} Y_{\ell m}^*(0, 0) = 2\pi \delta_{m,0} \sqrt{\frac{2\ell + 1}{4\pi}} P_\ell(0), \quad (14)$$

where we used that for $m = 0$

$$Y_{\ell 0}(\phi, \theta) = \sqrt{\frac{2\ell + 1}{4\pi}} P_\ell(\cos\theta). \quad (15)$$

Combining everything, we obtain the desired result

$$A_{lm} = \sqrt{\frac{4\pi}{2\ell + 1}} \frac{2QR^\ell}{\ell + 2} \delta_{m,0} P_\ell(0). \quad (16)$$

Plucking this back into the general formula (1) for the potential, we get for $r > R$

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} A_{\ell m} \frac{Y_{\ell m}(\phi, \theta)}{r^{\ell+1}} \quad (17)$$

$$= \frac{Q}{4\pi\epsilon_0} \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \sqrt{\frac{4\pi}{2\ell+1}} \frac{2R^{\ell}}{\ell+2} \delta_{m,0} P_{\ell}(0) \frac{Y_{\ell m}(\phi, \theta)}{r^{\ell+1}} \quad (18)$$

$$= \frac{Q}{4\pi\epsilon_0 r} \sum_{\ell=0}^{\infty} \left(\frac{R}{r}\right)^{\ell} \frac{2}{\ell+2} P_{\ell}(0) P_{\ell}(\cos\theta), \quad (19)$$

using again Eq. (15).

(1.b) Along the z -axis, we have $r = |z|$ and the potential is given by

$$\phi(z) = \frac{Q}{4\pi\epsilon_0 |z|} \sum_{\ell=0}^{\infty} \left(\frac{R}{|z|}\right)^{\ell} \frac{2}{\ell+2} P_{\ell}(0) \underbrace{P_{\ell}(1)}_{=1}. \quad (20)$$

In order to use the hint, we rewrite

$$\sum_{\ell=0}^{\infty} \left(\frac{R}{|z|}\right)^{\ell} \frac{1}{\ell+2} P_{\ell}(0) = \left(\frac{|z|}{R}\right)^2 \sum_{\ell=0}^{\infty} P_{\ell}(0) \int_0^{R/|z|} ds s^{\ell+1}. \quad (21)$$

We thus get

$$\phi(z) = \frac{2Q}{4\pi\epsilon_0 |z|} \left(\frac{z}{R}\right)^2 \int_0^{R/|z|} ds \sum_{\ell=0}^{\infty} P_{\ell}(0) s^{\ell+1} \quad (22)$$

$$= \frac{Q|z|}{2\pi\epsilon_0 R^2} \int_0^{R/|z|} ds \frac{s}{\sqrt{1+s^2}} \quad (23)$$

$$= \frac{Q|z|}{2\pi\epsilon_0 R^2} \left(\sqrt{1+(R/|z|)^2} - 1 \right) \quad (24)$$

$$= \frac{Q}{2\pi\epsilon_0 R^2} \left(\sqrt{z^2 + R^2} - |z| \right). \quad (25)$$

(1.c) Now, we look at the in-plane field, for which $z = 0 \Rightarrow \theta = \pi/2$.

$$\phi(r) \Big|_{z=0} = \frac{Q}{4\pi\epsilon_0 r} \sum_{\ell=0}^{\infty} \left(\frac{R}{r}\right)^{\ell} \frac{2}{\ell+2} P_{\ell}^2(0). \quad (26)$$

Using Hilb's formula, we find that asymptotically for $\ell \rightarrow \infty$

$$P_{\ell}^2(0) = \frac{2}{\pi\ell} \cos^2(\ell\pi/2) + \mathcal{O}(\ell^{-3/2}) = \frac{2}{\pi\ell} \begin{cases} 1 & \text{if } \ell \text{ is even} \\ 0 & \text{if } \ell \text{ is odd} \end{cases} + \mathcal{O}(\ell^{-3/2}). \quad (27)$$

Now we split the sum over angular momenta ℓ into two parts, summing the small values of ℓ which will always be finite, and then the large values of ℓ for which the expansion above

holds. We get

$$\phi(r) = \frac{Q}{4\pi\epsilon_0 r} \sum_{\ell \gg 1}^{\infty} \left(\frac{R}{r}\right)^{2\ell} \frac{2}{2\ell+2} \frac{2}{2\pi\ell} + \text{finite part} \quad (28)$$

$$= \frac{Q}{4\pi^2\epsilon_0 r} \sum_{\ell \gg 1}^{\infty} \left(\frac{R}{r}\right)^{2\ell} \frac{1}{\ell(\ell+1)} + \text{finite part} . \quad (29)$$

The only non-vanishing component of the field is the radial one given by

$$E_r = -\partial_r \phi \quad (30)$$

$$= -\partial_r \frac{Q}{4\pi^2\epsilon_0} \sum_{\ell \gg 1}^{\infty} \frac{R^{2\ell}}{r^{2\ell+1}} \frac{1}{\ell(\ell+1)} + \text{finite part} \quad (31)$$

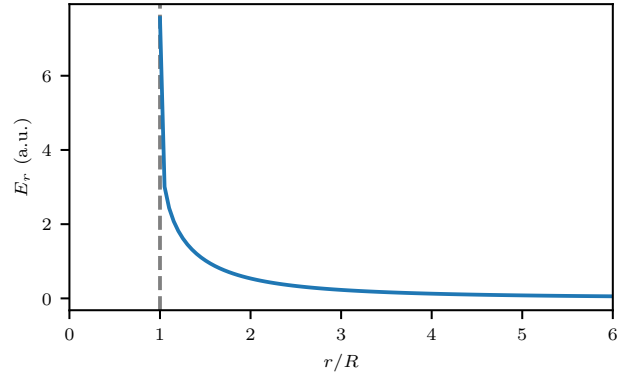
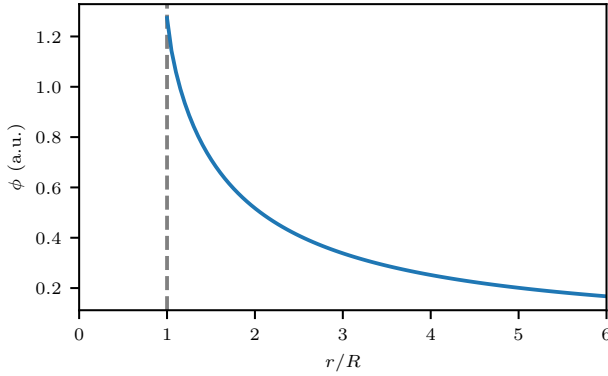
$$= \frac{Q}{4\pi^2\epsilon_0} \sum_{\ell \gg 1}^{\infty} \frac{R^{2\ell}}{r^{2\ell+2}} (2\ell+2) \frac{1}{\ell(\ell+1)} + \text{finite part} \quad (32)$$

$$= \frac{Q}{2\pi^2\epsilon_0 r^2} \sum_{\ell \gg 1}^{\infty} \left(\frac{R}{r}\right)^{2\ell} \frac{1}{\ell} + \text{finite part} \quad (33)$$

$$= -\frac{Q}{2\pi^2\epsilon_0 r^2} \log \left(1 - \left(\frac{R}{r}\right)^2 \right) + \text{finite part} \quad (34)$$

This clearly shows the logarithmic divergence as $r \searrow R$. Here we have used the series expansion of the logarithm

$$\log(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n} . \quad (35)$$



(a) The electric potential $\phi(\mathbf{r})$ for $r > R$ in the $x-y$ plane of the disk. (b) The radial component of the electric field $E_r(\mathbf{r})$ for $r > R$ in the $x-y$ plane of the disk.

(1.d) Here we plot the numeric calculation of the electric potential and electric field in the $x-y$ plane of the disk. Note that while the potential $\phi(r)$ stays finite near $r \gtrsim R$, the electric field indeed diverges. In the long distance limit, we see that $\phi(r)$ decays as $1/r$ and the field $E(r)$ decays as $1/r^2$ as expected.

Problem 2 The Electric Dipole

In this exercise, we consider two charges q and $-q$, separated by distance $a > 0$.

(2.a) First show that the electrostatic potential $\phi(\mathbf{r})$ of the dipole in the far-field region $|\mathbf{r}| \gg a$, to first nontrivial order in $\frac{|\mathbf{r}|}{a}$, is given by

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \mathbf{r}}{|\mathbf{r}|^3}, \quad \text{where } \mathbf{p} = qa \mathbf{e}_x. \quad (36)$$

(2.b) Using this, calculate the electric dipole field $\mathbf{E}(\mathbf{r})$ away from the origin, explicitly making use of $\mathbf{r} \neq 0$.

(2.c) Now show that the electric dipole field has a singular contribution $\frac{1}{3\epsilon_0} \mathbf{p} \delta(\mathbf{r})$ at $\mathbf{r} = 0$. To do so, integrate the electric field over a suitable volume $V \subset \mathbb{R}^3$ containing the dipole, and use Gauß's theorem as well as the relation $\mathbf{E} = -\nabla\phi$.

(2.d) Show that

$$dW = -\mathbf{E}(\mathbf{r}) \cdot d\mathbf{p} \quad (37)$$

is the infinitesimal work required to assemble a point electric dipole with dipole moment $d\mathbf{p}$ at position $\mathbf{r}$, assuming the charge used to build the dipole is initially dispersed at infinity. Here, $\mathbf{E}(\mathbf{r})$ denotes the electric field already present at the position $\mathbf{r}$, produced by all charges that have been assembled previously, but excluding the infinitesimal dipole itself. Thus, dW is the work needed to place the additional infinitesimal dipole in a pre-existing external field.

Solution:

(2.a)

$$\begin{aligned} \rho(\mathbf{r}) &= q \left(\delta(\mathbf{r} - (0, 0, a/2)^T) - \delta(\mathbf{r} + (0, 0, a/2)^T) \right), \\ \phi(\mathbf{r}) &= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{\|\mathbf{r} - (0, 0, a/2)^T\|} - \frac{1}{\|\mathbf{r} + (0, 0, a/2)^T\|} \right). \end{aligned}$$

For $a \ll \|\mathbf{r}\|$, one may use a Taylor expansion about $a = 0$.

$$\phi^{(0)}(\mathbf{r}) = \frac{q}{4\pi\epsilon_0} \left(\frac{1}{\|\mathbf{r} - (0, 0, a/2)^T\|} - \frac{1}{\|\mathbf{r} + (0, 0, a/2)^T\|} \right) \Big|_{a=0} = 0.$$

To first order:

$$\begin{aligned}
\phi^{(1)}(\mathbf{r}) &= \frac{q}{4\pi\epsilon_0} \left(a \frac{\partial}{\partial a} \frac{1}{\|\mathbf{r} - (0, 0, a/2)^\top\|} \Big|_{a=0} - a \frac{\partial}{\partial a} \frac{1}{\|\mathbf{r} + (0, 0, a/2)^\top\|} \Big|_{a=0} \right) \\
&= \frac{q}{4\pi\epsilon_0} a \frac{\partial}{\partial a} \left(\frac{1}{\sqrt{x^2 + y^2 + (z - a/2)^2}} - \frac{1}{\sqrt{x^2 + y^2 + (z + a/2)^2}} \right) \Big|_{a=0} \\
&= \frac{q}{4\pi\epsilon_0} a \left(\frac{z - a/2}{2(x^2 + y^2 + (z - a/2)^2)^{3/2}} \Big|_{a=0} + \frac{z + a/2}{2(x^2 + y^2 + (z + a/2)^2)^{3/2}} \Big|_{a=0} \right) \\
&= \frac{q}{4\pi\epsilon_0} a \left(\frac{1}{2} \frac{z}{\|\mathbf{r}\|^3} + \frac{1}{2} \frac{z}{\|\mathbf{r}\|^3} \right) = \frac{qa}{4\pi\epsilon_0} \frac{z}{\|\mathbf{r}\|^3}.
\end{aligned}$$

With $\mathbf{p} = qa \mathbf{e}_z$ and $z = \mathbf{r} \cdot \mathbf{e}_z$, this becomes

$$\phi(\mathbf{r}) \approx \phi^{(0)}(\mathbf{r}) + \phi^{(1)}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \mathbf{r}}{\|\mathbf{r}\|^3}, \quad a \ll \|\mathbf{r}\|.$$

(2.b)

$$\mathbf{E}(\mathbf{r}) = -\nabla\phi(\mathbf{r}).$$

Spherical coordinates parametrize $\mathbb{R}^3 \setminus \{0\}$. From the literature (e.g. Griffiths, p. 42), one has for ∇ in spherical coordinates (r, θ, ϕ) :

$$\nabla = \hat{\mathbf{r}} \frac{\partial}{\partial r} + \hat{\boldsymbol{\theta}} \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{\boldsymbol{\phi}} \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}.$$

Since the two point charges lie on the z -axis, we have

$$\mathbf{p} = qa \mathbf{e}_z, \quad \mathbf{p} \cdot \mathbf{r} = qa r \cos \theta.$$

Hence

$$\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \frac{\mathbf{p} \cdot \mathbf{r}}{\|\mathbf{r}\|^3} = \frac{1}{4\pi\epsilon_0} \frac{qa \cos \theta}{r^2}.$$

Therefore:

$$\begin{aligned}
4\pi\epsilon_0 \nabla\phi(\mathbf{r}) &= \hat{\mathbf{r}} \frac{\partial}{\partial r} \frac{qa \cos \theta}{r^2} + \hat{\boldsymbol{\theta}} \frac{1}{r} \frac{\partial}{\partial \theta} \frac{qa \cos \theta}{r^2} + \hat{\boldsymbol{\phi}} \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \frac{qa \cos \theta}{r^2} \\
&= -2\hat{\mathbf{r}} \frac{qa \cos \theta}{r^3} - \hat{\boldsymbol{\theta}} \frac{qa \sin \theta}{r^3}.
\end{aligned}$$

Thus

$$\mathbf{E}(\mathbf{r}) = -\nabla\phi(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left(2\hat{\mathbf{r}} \frac{qa \cos \theta}{r^3} + \hat{\boldsymbol{\theta}} \frac{qa \sin \theta}{r^3} \right). \quad (*)$$

Clearly $\hat{\phi} \cdot \mathbf{p} = 0$, i.e. $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\theta}}$, and $\mathbf{p}$ are coplanar. In particular,

$$\begin{aligned}\mathbf{p} &= (\mathbf{p} \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} + (\mathbf{p} \cdot \hat{\boldsymbol{\theta}}) \hat{\boldsymbol{\theta}} \\ &= \|\mathbf{p}\| \cos \theta \hat{\mathbf{r}} - \|\mathbf{p}\| \sin \theta \hat{\boldsymbol{\theta}}.\end{aligned}$$

Hence (*) becomes

$$\begin{aligned}\mathbf{E}(\mathbf{r}) &= \frac{1}{4\pi\epsilon_0} \left(3\hat{\mathbf{r}} \frac{qa \cos \theta}{r^3} - \left(\frac{qa \cos \theta}{r^3} \hat{\mathbf{r}} - \frac{qa \sin \theta}{r^3} \hat{\boldsymbol{\theta}} \right) \right) \\ &= \frac{1}{4\pi\epsilon_0} \left(3 \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^3} \hat{\mathbf{r}} - \frac{\mathbf{p}}{r^3} \right).\end{aligned}$$

(2.c)

$$\begin{aligned}B_\epsilon &:= \{\mathbf{r} \in \mathbb{R}^3 : \|\mathbf{r}\| < \epsilon\}, \\ S_\epsilon &:= \partial B_\epsilon = \{\mathbf{r} \in \mathbb{R}^3 : \|\mathbf{r}\| = \epsilon\}.\end{aligned}$$

Consider $\int_{B_\epsilon} \mathbf{E}(\mathbf{r}) d^3\mathbf{r}$ in its i -th component ($i \in \{1, 2, 3\}$):

$$\begin{aligned}- \left(\int_{B_\epsilon} \mathbf{E}(\mathbf{r}) d^3\mathbf{r} \right)_i &= -\mathbf{e}_i \cdot \int_{B_\epsilon} \mathbf{E}(\mathbf{r}) d^3\mathbf{r} = - \int_{B_\epsilon} \mathbf{e}_i \cdot \mathbf{E}(\mathbf{r}) d^3\mathbf{r} \\ &= \int_{B_\epsilon} \mathbf{e}_i \cdot \nabla \phi(\mathbf{r}) d^3\mathbf{r} = \int_{B_\epsilon} \nabla \cdot (\mathbf{e}_i \phi(\mathbf{r})) d^3\mathbf{r} \\ &\stackrel{\text{Gauss}}{=} \int_{S_\epsilon} (\mathbf{e}_i \phi(\mathbf{r})) \cdot \hat{\mathbf{n}} dA = \mathbf{e}_i \cdot \int_{S_\epsilon} \phi(\mathbf{r}) \hat{\mathbf{n}} dA.\end{aligned}$$

Since i is arbitrary, we may also write:

$$\int_{B_\epsilon} \mathbf{E}(\mathbf{r}) d^3\mathbf{r} = - \int_{S_\epsilon} \phi(\mathbf{r}) \hat{\mathbf{n}} dA. \quad (**)$$

In spherical coordinates,

$$\hat{\mathbf{n}} = \hat{\mathbf{r}}, \quad dA = \epsilon^2 \sin \theta d\theta d\phi, \quad \theta \in [0, \pi], \quad \phi \in [0, 2\pi].$$

Thus:

$$\begin{aligned}(**) &= - \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta \epsilon^2 \frac{1}{4\pi\epsilon_0} \frac{qa \cos \theta}{\epsilon^2} \hat{\mathbf{r}} \\ &= - \frac{qa}{4\pi\epsilon_0} \int_0^{2\pi} d\phi \int_0^\pi d\theta \sin \theta \cos \theta \begin{pmatrix} \sin \theta \cos \phi \\ \sin \theta \sin \phi \\ \cos \theta \end{pmatrix} \\ &= - \frac{qa}{4\pi\epsilon_0} \mathbf{e}_z 2\pi \int_0^\pi \sin \theta \cos^2 \theta d\theta \\ &= - \frac{qa}{2\epsilon_0} \mathbf{e}_z \int_{-1}^1 \eta^2 d\eta = - \frac{\mathbf{p}}{6\epsilon_0} [\eta^3]_{-1}^1 = - \frac{\mathbf{p}}{3\epsilon_0}.\end{aligned}$$

On the other hand,

$$\int_{B_\varepsilon} -\frac{\mathbf{p}}{3\varepsilon_0} \delta(\mathbf{r}) d^3\mathbf{r} = -\frac{\mathbf{p}}{3\varepsilon_0}.$$

Since the integrals over arbitrary small balls agree, it follows in the distributional sense that

$$\mathbf{E}(\mathbf{r}) = -\frac{\mathbf{p}}{3\varepsilon_0} \delta(\mathbf{r}) + \frac{1}{4\pi\varepsilon_0} \left(3 \frac{\mathbf{p} \cdot \hat{\mathbf{r}}}{r^3} \hat{\mathbf{r}} - \frac{\mathbf{p}}{r^3} \right).$$

(2.d) Let $\mathbf{E}(\mathbf{r})$ be the external electric field at point $\mathbf{r}$.

To build the infinitesimal dipole, the charges $\pm dq$ at

$$\mathbf{r}_\pm = \mathbf{r} \pm \frac{\boldsymbol{\ell}}{2} \quad (38)$$

have to be assembled. $\boldsymbol{\ell}$ points from $+dq$ to $-dq$.

Assuming that the potential $\phi(\mathbf{x})$ associated with $\mathbf{E}(\mathbf{r})$ vanishes at infinity, the work needed to bring the charges to their positions is

$$dW = dq \phi(\mathbf{r}_+) - dq \phi(\mathbf{r}_-) = dq (\phi(\mathbf{r}_+) - \phi(\mathbf{r}_-)). \quad (39)$$

Expanding about $\mathbf{r}$,

$$\phi\left(\mathbf{r} \pm \frac{\boldsymbol{\ell}}{2}\right) = \phi(\mathbf{r}) \pm \frac{1}{2} \boldsymbol{\ell} \cdot \boldsymbol{\nabla} \phi(\mathbf{r}) + \mathcal{O}(\ell^2). \quad (40)$$

$$\Rightarrow dW = dq \boldsymbol{\ell} \cdot \boldsymbol{\nabla} \phi(\mathbf{r}) + \mathcal{O}(\ell^2) = -d\mathbf{p} \cdot \mathbf{E}(\mathbf{r}) + \mathcal{O}(\ell^2). \quad (41)$$

Problem 3 A charged rod in front of a conducting plane

The diagram below shows an uniformly charged thin rod of length L and net charge Q , oriented parallel to a grounded infinite conducting plane at the distance d from the plane.

- (3.a) Compute the electrostatic potential of the configuration by integrating over a suitable Dirichlet Green's function, and confirm that the boundary condition on the plane is fulfilled.
- (3.b) Find the force exerted on the rod by the plane.
- (3.c) Simplify your result in part (b) in the limit $d \gg L$. Can you give a physical intuition for your result?
- (3.d) Find the charge density $\sigma(x, y)$ induced on the conducting plane by the rod.
- (3.e) Find the total charge induced on the plane *without* integrating $\sigma(x, y)$.

Solution:

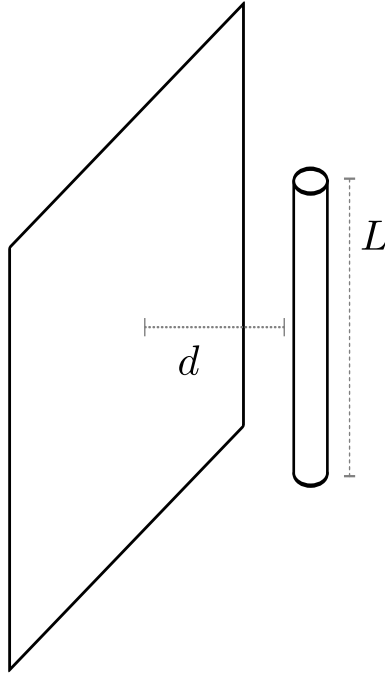


Abbildung 2: Charged rod of length L at distance d from perfectly conducting plane.

- (3.a) Let the grounded plane be the x - y plane. Let the rod lie along the x -axis at $z = d$, $y = 0$. Its linear charge density is

$$\lambda = \frac{Q}{L}.$$

The Dirichlet Green's function for the infinite conducting plane is

$$G_D = \frac{1}{4\pi\epsilon_0} \left(\frac{1}{\|\mathbf{r} - \mathbf{r}'\|} - \frac{1}{\|\mathbf{r} - \mathbf{r}'^*\|} \right),$$

with

$$\mathbf{r}'^* = (x', y', -z').$$

The charge density of the rod is

$$\rho(\mathbf{r}') = \lambda \delta(y') \delta(z' - d) \Theta\left(\frac{L}{2} - |x'|\right).$$

Hence

$$\begin{aligned} \phi(\mathbf{r})|_{z \geq 0} &= \int_{z' > 0} G_D(\mathbf{r}; \mathbf{r}') \rho(\mathbf{r}') d^3\mathbf{r}' \\ &= \frac{\lambda}{4\pi\epsilon_0} \int_{-L/2}^{+L/2} dx' \left[\frac{1}{\sqrt{(x - x')^2 + y^2 + (z - d)^2}} - \frac{1}{\sqrt{(x - x')^2 + y^2 + (z + d)^2}} \right] \\ &= \frac{\lambda}{4\pi\epsilon_0} \left[\operatorname{arsinh}\left(\frac{x + \frac{L}{2}}{\sqrt{(z - d)^2 + y^2}}\right) - \operatorname{arsinh}\left(\frac{x - \frac{L}{2}}{\sqrt{(z - d)^2 + y^2}}\right) \right. \\ &\quad \left. - \operatorname{arsinh}\left(\frac{x + \frac{L}{2}}{\sqrt{(z + d)^2 + y^2}}\right) + \operatorname{arsinh}\left(\frac{x - \frac{L}{2}}{\sqrt{(z + d)^2 + y^2}}\right) \right]. \end{aligned}$$

This vanishes at $z = 0$, since

$$\sqrt{(z+d)^2 + y^2} \Big|_{z=0} = \sqrt{(z-d)^2 + y^2} \Big|_{z=0}.$$

(3.b) The z -component of the electric field due to the plane is

$$E_z^{\text{plane}}(x, 0, d) = - \frac{\partial \phi^{\text{im}}}{\partial z} \Big|_{y=0, z=d},$$

with ϕ^{im} the image-charge contribution to ϕ .

Thus

$$\begin{aligned} E_z^{\text{plane}}(x, 0, d) &= -\frac{\lambda}{4\pi\epsilon_0} \frac{\partial}{\partial z} \left(-\text{arsinh} \left(\frac{x + \frac{L}{2}}{\sqrt{(z+d)^2 + y^2}} \right) + \text{arsinh} \left(\frac{x - \frac{L}{2}}{\sqrt{(z+d)^2 + y^2}} \right) \right) \\ &= -\frac{\lambda}{4\pi\epsilon_0} \left[\frac{x + \frac{L}{2}}{2d \sqrt{4d^2 + (x + \frac{L}{2})^2}} - \frac{x - \frac{L}{2}}{2d \sqrt{4d^2 + (x - \frac{L}{2})^2}} \right]. \end{aligned}$$

The total force is then ($F_x, F_y = 0$ from symmetry)

$$\begin{aligned} F_z &= \int_{-L/2}^{+L/2} \lambda E_z^{\text{plane}} dx \\ &= -\frac{\lambda^2}{4\pi\epsilon_0 d} \left(\sqrt{L^2 + 4d^2} - 2d \right). \end{aligned}$$

(3.c) For $d \gg L$, we write

$$\sqrt{L^2 + 4d^2} = 2d + \frac{L^2}{4d} + \mathcal{O}\left(\frac{L^4}{d^3}\right).$$

Hence

$$F_z = -\frac{Q^2}{16\pi\epsilon_0 d^2} = -\frac{Q^2}{4\pi\epsilon_0 (2d)^2},$$

and the rod behaves like a point charge for $d \gg L$.

(3.d) For conductors we have

$$\mathbf{E} = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}} \quad \text{at the boundary.}$$

Hence

$$|\mathbf{E}| = \frac{\sigma}{\epsilon_0}.$$

We need to evaluate

$$E_{\perp} = - \frac{\partial \phi}{\partial z} \Big|_{z=0},$$

but now with ϕ being the full potential from (a).

We find that

$$E_{\perp} = -\frac{2\lambda}{4\pi\epsilon_0} \left[\frac{(x + \frac{L}{2})d}{(y^2 + d^2)\sqrt{4d^2 + (x + \frac{L}{2})^2}} - \frac{(x - \frac{L}{2})d}{(y^2 + d^2)\sqrt{4d^2 + (x - \frac{L}{2})^2}} \right].$$

Therefore

$$\sigma(x, y) = -\frac{\lambda}{2\pi} \frac{d}{y^2 + d^2} \left[\frac{x + \frac{L}{2}}{\sqrt{4d^2 + (x + \frac{L}{2})^2}} - \frac{x - \frac{L}{2}}{\sqrt{4d^2 + (x - \frac{L}{2})^2}} \right].$$

- (3.e) Use Gauß's law and take a very large closed surface. Choose one boundary of that surface to lie on the plane. Inside the conductor, we have $\mathbf{E} = 0$, hence there is no flux through that part of the boundary.

Close the surface in the half-space $z > 0$, sufficiently far away from the rod. Then the enclosed charge is

$$Q_{\text{tot}} = Q_{\text{ind}} + Q.$$

But the field is that of a real rod with charge $+Q$ and an image rod with charge $-Q$. The total charge of this equivalent system is 0, hence

$$Q_{\text{tot}} = 0 \quad \text{and} \quad Q_{\text{ind}} = -Q.$$